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CECS 327 - Project 2

Task 1: Anycast with Docker (TCP)

Example outputs:
`docker-compose up –build`

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

 => => unpacking to docker.io/library/task1-anycast-with-docker-server2:latest                                    0.2s
 => [server1] exporting to image                                                                                  0.9s
 => => exporting layers                                                                                           0.6s
 => => exporting manifest sha256:832439cdb369140673d3e787cf2629ea7a73352cc177c8c61897eefffa8b5895                 0.0s
 => => exporting config sha256:c6438271f4dd19ec2da2916fce79dec1f639c161105cda36f8ce518932e3511b                   0.0s
 => => exporting attestation manifest sha256:c9fac0fb780cd213c3f4b4d6deac34e876b21e6700620235898 97ab0cfcb7559     0.0s
 => => exporting manifest list sha256:7e617970f94ace04d2d5c22a8c8421454bc893935b21881816a9206336f9684d            0.0s
 => => naming to docker.io/library/task1-anycast-with-docker-server1:latest                                       0.0s
 => => unpacking to docker.io/library/task1-anycast-with-docker-server1:latest                                    0.2s
 => [client] resolving provenance for metadata file                                                               0.0s
 => [server1] resolving provenance for metadata file                                                              0.0s
 => [server3] resolving provenance for metadata file                                                              0.0s
 => [server2] resolving provenance for metadata file                                                              0.0s
[+] up 9/9
 ✔ Image task1-anycast-with-docker-server3       Built                                                            5.3s
 ✔ Image task1-anycast-with-docker-client        Built                                                            5.3s
 ✔ Image task1-anycast-with-docker-server1       Built                                                            5.3s
 ✔ Image task1-anycast-with-docker-server2       Built                                                            5.3s
 ✔ Network task1-anycast-with-docker_default     Created                                                          0.0s
 ✔ Container task1-anycast-with-docker-server2-1 Created                                                          0.1s
 ✔ Container task1-anycast-with-docker-server1-1 Created                                                          0.1s
 ✔ Container task1-anycast-with-docker-server3-1 Created                                                          0.1s
 ✔ Container task1-anycast-with-docker-client-1  Created                                                          0.1s
Attaching to client-1, server1-1, server2-1, server3-1
server2-1   | Server ready on port 5000
server3-1   | Server ready on port 5000
server1-1   | Server ready on port 5000
server1-1   | Accepted connection from ('172.19.0.5', 55644)
server1-1   | Sent: Hello from server1 !!!


server3-1   | Accepted connection from ('172.19.0.5', 59368)
server2-1   | Accepted connection from ('172.19.0.5', 54068)



server3-1   | Sent: Hello from server3 !!!
client-1    | Received: Hello from server1 !!!
server2-1   | Sent: Hello from server2 !!!


client-1    | Received: Hello from server2 !!!
client-1    | Received: Hello from server3 !!!
client-1 exited with code 0
server1-1   | Accepted connection from ('172.19.0.5', 60520)
server1-1   | Sent: Hello from server1 !!!

server2-1   | Accepted connection from ('172.19.0.5', 43044)
server3-1   | Accepted connection from ('172.19.0.5', 36540)ed: Hello from server1 !!!
task1-anycast-with-docker-client-run-395f4c01a62f   | Received: Hello from server2 !!!


server2-1                                           | Sent: Hello from server2 !!!
server3-1                                           | Sent: Hello from server3 !!!
task1-anycast-with-docker-client-run-395f4c01a62f exited with code 0

v View in Docker Desktop   o View Config   w Enable Watch   d Detach
```

`tcpdump` logs

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

❯ docker exec -it task1-anycast-with-docker-server1-1 tcpdump -i eth0 port 5000
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
20:16:26.973712 IP task1-anycast-with-docker-client-run-395f4c01a62f.task1-anycast-with-docker_default.60520 > d1ab09d5bb09.5000: Flags [S], seq 1382425849, win 64240, options [mss 1460,sa
ckOK,TS val 1609978971 ecr 0,nop,wscale 7], length 0
20:16:26.973732 IP d1ab09d5bb09.5000 > task1-anycast-with-docker-client-run-395f4c01a62f.task1-anycast-with-docker_default.60520: Flags [S.], seq 4044959396, ack 1382425850, win 65160, opt
ions [mss 1460,sackOK,TS val 3767662359 ecr 1609978971,nop,wscale 7], length 0
20:16:26.973746 IP task1-anycast-with-docker-client-run-395f4c01a62f.task1-anycast-with-docker_default.60520 > d1ab09d5bb09.5000: Flags [.], ack 1, win 502, options [nop,nop,TS val 1609978
971 ecr 3767662359], length 0
20:16:26.973828 IP d1ab09d5bb09.5000 > task1-anycast-with-docker-client-run-395f4c01a62f.task1-anycast-with-docker_default.60520: Flags [P.], seq 1:24, ack 1, win 510, options [nop,nop,TS
val 3767662359 ecr 1609978971], length 23
20:16:26.973840 IP task1-anycast-with-docker-client-run-395f4c01a62f.task1-anycast-with-docker_default.60520 > d1ab09d5bb09.5000: Flags [.], ack 24, win 502, options [nop,nop,TS val 160997
8971 ecr 3767662359], length 0
20:16:26.973849 IP d1ab09d5bb09.5000 > task1-anycast-with-docker-client-run-395f4c01a62f.task1-anycast-with-docker_default.60520: Flags [F.], seq 24, ack 1, win 510, options [nop,nop,TS va
l 3767662359 ecr 1609978971], length 0
20:16:26.973882 IP task1-anycast-with-docker-client-run-395f4c01a62f.task1-anycast-with-docker_default.60520 > d1ab09d5bb09.5000: Flags [F.], seq 1, ack 25, win 502, options [nop,nop,TS va
l 1609978971 ecr 3767662359], length 0
20:16:26.973885 IP d1ab09d5bb09.5000 > task1-anycast-with-docker-client-run-395f4c01a62f.task1-anycast-with-docker_default.60520: Flags [.], ack 2, win 510, options [nop,nop,TS val 3767662
359 ecr 1609978971], length 0
```

`docker-compose run client`

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

❯ ls
README.md                    proj1-docker-containers        proj2-distributed-communication
❯ cd proj2
cd: no such file or directory: proj2
❯ cd proj2-distributed-communication
❯ ls
task1-anycast-with-docker
❯ cd task1-anycast-with-docker
❯ ls
Dockerfile              client.py              docker-compose.yml      server.py
❯ docker-compose run client
[+]  3/3t 3/33
 ✔ Container task1-anycast-with-docker-server2-1 Running                                    0.0s
 ✔ Container task1-anycast-with-docker-server1-1 Running                                    0.0s
 ✔ Container task1-anycast-with-docker-server3-1 Running                                    0.0s
Container task1-anycast-with-docker-client-run-395f4c01a62f Creating
Container task1-anycast-with-docker-client-run-395f4c01a62f Created
Received: Hello from server1 !!!
Received: Hello from server2 !!!
Received: Hello from server3 !!!

~/s/e/u/sp26/32/327_Docker/proj2/task1-anycast-with-docker on main !2 ····· at 13:16:27
❯
```

Task 2: Multicast with UDP

Example outputs:

`docker-compose up –build`

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

❯ clear
❯ docker-compose up
[+] up 5/5
 ✔ Network task2-multicast-with-udp_default       Created                                   0.0s
 ✔ Container task2-multicast-with-udp-sender2-1   Created                                   0.0s
 ✔ Container task2-multicast-with-udp-sender1-1   Created                                   0.0s
 ✔ Container task2-multicast-with-udp-receiver1-1 Created                                   0.0s
 ✔ Container task2-multicast-with-udp-receiver2-1 Created                                   0.0s
Attaching to receiver1-1, receiver2-1, sender1-1, sender2-1
sender1-1   | Sender started — broadcasting as sensor: temp
sender1-1   | Sent: {"sensor": "temp", "value": 29.8}
sender1-1   | Sent binary data: b'\x7f\x8b\x1c\xb4\n\x03\xe2\xd9'
sender2-1   | Sender started — broadcasting as sensor: humidity
sender2-1   | Sent: {"sensor": "humidity", "value": 25.5}
sender2-1   | Sent binary data: b'\x84%\xf3\xd1T\x92\x86\x91'
receiver2-1 | Joined multicast group
receiver1-1 | Joined multicast group

receiver1-1 | Received: {'sensor': 'temp', 'value': 27.6} from ('172.19.0.2', 60319)

receiver1-1 | Received binary data from ('172.19.0.2', 60319): b'\x06d\xc6\x9d\x81X\xac('
receiver2-1 | Received binary data from ('172.19.0.2', 60319): b'\x06d\xc6\x9d\x81X\xac('
receiver2-1 | Received: {'sensor': 'humidity', 'value': 20.7} from ('172.19.0.3', 59796)

receiver1-1 | Received: {'sensor': 'humidity', 'value': 20.7} from ('172.19.0.3', 59796)
sender2-1   | Sent: {"sensor": "humidity", "value": 20.7}
receiver2-1 | Received binary data from ('172.19.0.3', 59796): b']UZ\xaa\xa8\x08oY'

sender2-1   | Sent binary data: b']UZ\xaa\xa8\x08oY'atch    d Detach

receiver1-1 | Received binary data from ('172.19.0.3', 59796): b']UZ\xaa\xa8\x08oY'

sender1-1   | Sent: {"sensor": "temp", "value": 27.7}
receiver2-1 | Received: {'sensor': 'temp', 'value': 27.7} from ('172.19.0.2', 60319)
receiver1-1 | Received: {'sensor': 'temp', 'value': 27.7} from ('172.19.0.2', 60319)

receiver1-1 | Received binary data from ('172.19.0.2', 60319): b'c\xee\x8b\x88\x9f\xebtS'
receiver2-1 | Received binary data from ('172.19.0.2', 60319): b'c\xee\x8b\x88\x9f\xebtS'

receiver1-1 | Received: {'sensor': 'humidity', 'value': 26.7} from ('172.19.0.3', 59796)
sender2-1   | Sent: {"sensor": "humidity", "value": 26.7}

Ln 11, Col 1   Spaces: 4   UTF-8   LF   { } Python   3.13.7   Python 3.14.3 (homebrew)
```

`tcpdump` logs

```
❯ docker exec -it task2-multicast-with-udp-receiver1-1 tcpdump -i eth0 udp port 5007
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
23:28:07.073599 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 33
23:28:07.073599 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 37
23:28:07.073677 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 8
23:28:07.073771 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 8
23:28:08.077126 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 33
23:28:08.077278 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 8
23:28:08.077815 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 37
23:28:08.077908 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 8
23:28:09.082122 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 37
23:28:09.082122 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 33
23:28:09.082172 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 8
23:28:09.082290 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 8
23:28:10.086629 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 33
23:28:10.086772 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 8
23:28:10.087213 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 37
23:28:10.087299 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 8
23:28:11.088071 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 33
23:28:11.088086 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 37
23:28:11.088164 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 8
23:28:11.088174 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 8
23:28:12.092209 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 33
23:28:12.092361 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 8
23:28:12.092505 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 37
23:28:12.092686 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 8
23:28:13.094945 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 33
23:28:13.094946 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 37
23:28:13.095082 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 8
23:28:13.095082 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 8
23:28:14.096833 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 37
23:28:14.096833 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 33
23:28:14.096891 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 8
23:28:14.096891 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 8
23:28:15.103269 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 37
23:28:15.103268 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 33
23:28:15.103433 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 8
23:28:15.103439 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 8
23:28:16.106229 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 37
23:28:16.106229 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 33
23:28:16.106560 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 8
23:28:16.106560 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 8
```

Docker logs:

`docker logs task2-multicast-with-udp-receiver1-1`

```
❯ docker logs task2-multicast-with-udp-receiver1-1
Joined multicast group
Received: Multicast message from ('172.19.0.3', 44968)
Received: {'sensor': 'humidity', 'value': 29.2}
Received: Multicast message from ('172.19.0.2', 53790)
Received: {'sensor': 'temp', 'value': 27.1}
Received: Multicast message from ('172.19.0.3', 44968)
Received binary data from ('172.19.0.3', 44968): b'\xc4{l\xf9\xcf\x01\x02\xdf'
Received: Multicast message from ('172.19.0.2', 53790)
Received binary data from ('172.19.0.2', 53790): b'\x07B\xd7\xbe\x19\x96%\xd7'
Received: Multicast message from ('172.19.0.2', 53790)
Received: {'sensor': 'temp', 'value': 23.2}
Received: Multicast message from ('172.19.0.3', 44968)
Received: {'sensor': 'humidity', 'value': 25.9}
Received: Multicast message from ('172.19.0.3', 44968)
Received binary data from ('172.19.0.3', 44968): b'"\xfdB\xb9\xd8u\xee\xf8'
Received: Multicast message from ('172.19.0.2', 53790)
Received binary data from ('172.19.0.2', 53790): b'\xa87\x9b$\xaes\x83\x16'
Received: Multicast message from ('172.19.0.2', 53790)
Received: {'sensor': 'temp', 'value': 25.3}
Received: Multicast message from ('172.19.0.2', 53790)
Received binary data from ('172.19.0.2', 53790): b'\x9fy-[G\xcc\x18\xaf'
Received: Multicast message from ('172.19.0.3', 44968)
Received: {'sensor': 'humidity', 'value': 25.3}
Received: Multicast message from ('172.19.0.3', 44968)
Received binary data from ('172.19.0.3', 44968): b'\xe2\xaf\x93\x91\xa54\xf5\xd0'
Received: Multicast message from ('172.19.0.3', 44968)
Received: {'sensor': 'humidity', 'value': 28.1}
Received: Multicast message from ('172.19.0.2', 53790)
Received: {'sensor': 'temp', 'value': 23.9}
Received: Multicast message from ('172.19.0.3', 44968)
Received binary data from ('172.19.0.3', 44968): b'\x16\x03\x8fG\x9b\x14\x8f\x00'
Received: Multicast message from ('172.19.0.2', 53790)
Received binary data from ('172.19.0.2', 53790): b'\x8d\x8c!\xae}\xca\xbf\x9c'
Received: Multicast message from ('172.19.0.2', 53790)
Received: {'sensor': 'temp', 'value': 28.1}
Received: Multicast message from ('172.19.0.3', 44968)
Received: {'sensor': 'humidity', 'value': 22.5}
Received: Multicast message from ('172.19.0.3', 44968)
Received binary data from ('172.19.0.3', 44968): b'\xfeI\x01\xb0)\xfb\xed7'
Received: Multicast message from ('172.19.0.2', 53790)
Received binary data from ('172.19.0.2', 53790): b'w\xd7\xb1\xc9\xa8\xbe8\xb0'
Received: Multicast message from ('172.19.0.2', 53790)
Received: {'sensor': 'temp', 'value': 25.1}
Received: Multicast message from ('172.19.0.3', 44968)
Received: {'sensor': 'humidity', 'value': 27.1}
Received: Multicast message from ('172.19.0.2', 53790)
Received binary data from ('172.19.0.2', 53790): b'\xd3\x0e\xe9\x08\xb8\xd2\x8b|'
Received: Multicast message from ('172.19.0.3', 44968)
Received binary data from ('172.19.0.3', 44968): b'\x1a\xc9n\x1b\xe32K1'
Received: Multicast message from ('172.19.0.3', 44968)
Received: {'sensor': 'humidity', 'value': 23.5}
Received: Multicast message from ('172.19.0.2', 53790)
Received: {'sensor': 'temp', 'value': 20.2}
Received: Multicast message from ('172.19.0.3', 44968)
Received binary data from ('172.19.0.3', 44968): b";\xd5\x9b\x14\x0fn'c"
Received: Multicast message from ('172.19.0.2', 53790)
Received binary data from ('172.19.0.2', 53790): b'\x9ed|Qf\\]\xe3'
Received: Multicast message from ('172.19.0.2', 53790)
Received: {'sensor': 'temp', 'value': 24.1}
Received: Multicast message from ('172.19.0.2', 53790)
```

`docker logs task2-multicast-with-udp-sender2-1`

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

❯ docker logs task2-multicast-with-udp-sender2-1
Sender started — broadcasting as sensor: humidity
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 26.9}
Sent binary data: b'\x19c\x12\x97,\x02\x19\xda'
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 29.2}
Sent binary data: b'\xc4{l\xf9\xcf\x01\x02\xdf'
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 25.9}
Sent binary data: b'"\xfdB\xb9\xd8u\xee\xf8'
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 25.3}
Sent binary data: b'\xe2\xaf\x93\x91\xa54\xf5\xd0'
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 28.1}
Sent binary data: b'\x16\x03\x8fG\x9b\x14\x8f\x00'
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 22.5}
Sent binary data: b'\xfeI\x01\xb0)\xfb\xed7'
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 27.1}
Sent binary data: b'\x1a\xc9n\x1b\xe32K1'
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 23.5}
Sent binary data: b";\xd5\x9b\x14\x0fn'c"
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 22.4}
Sent binary data: b'L\x8e\x04\x1a\t\xf7\xc2r'
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 25.3}
Sent binary data: b'\x97u\x9f\x97\x17\xfb\xc9@'
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 23.3}
Sent binary data: b'p\xc2\xe9\xd0+\xb8\x0f\x19'
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 28.4}
Sent binary data: b'\xdb=\xad\xac\xda\xe6\xf9\x8a'
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 20.2}
Sent binary data: b'E\xba\xe7 \x1d\x85\xb4\xb0'
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 22.9}
Sent binary data: b'/|\x1c\xbf<\xd2\xaf\x87'
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 22.2}
Sent binary data: b'\x82y\xf6V\xcc\xce\xf8\xff'
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 27.0}
Sent binary data: b"F\x8c\xba'-\x1d:\x96"
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 21.3}
Sent binary data: b'#9\xfb\x9d\x82\x10\x9bh'
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 25.3}
Sent binary data: b'\x18\xf2\xc6r\t\x0e\x8co'
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 23.0}
Sent binary data: b'\xc6\r,h\x1dx\x19\x9e'
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 25.1}

zsh task2-...
zsh

Ln 26, Col 5   Spaces: 4   UTF-8   LF   { } Python   3.13.7   Python 3.14.3 (homebrew)
```

Section 1: Why UDP Multicast is Not Reliable

UDP is a connectionless, unreliable datagram service with no acknowledgments, retransmission, flow control, or ordering guarantees. When a sender broadcasts to 224.1.1.1, it has no knowledge of who received the packet; if the network drops it, neither side is notified.

There is no handshake and no recovery mechanism. Any ordering guarantees like FIFO or causal ordering would need to be built on top of the transport layer, raw IP multicast provides none of this. This was visible in our implementation: the receiver's 1-second timeout loop means any packet arriving at the wrong moment is silently missed, consistent with UDP's fire-and-forget model.

Section 2: How Group Join/Leave Affects Who Receives Packets

Only current group members receive multicast messages; the sender has no knowledge of who is listening. A receiver joins by registering IP_ADD_MEMBERSHIP with the OS, and leaves when its duration timer expires and the socket closes. There is no buffering for late joiners or departed members; packets sent before joining or after leaving are permanently lost. This was demonstrated directly in our project: once receiver1 and receiver2 hit their 60-second limit and printed "Leaving multicast group," they received nothing further despite the senders continuing to broadcast.

Section 3: Single-Source vs. Multi-Source Multicast

With a single source, all packets originate from one IP, making ordering and drop detection straightforward. In our multi-source setup, sender1 (172.19.0.2, temp) and sender2 (172.19.0.3, humidity) broadcast independently to the same group address. Receivers get interleaved messages from both with no coordination between senders; arrival order is determined purely by network timing. This was confirmed in both the receiver logs and tcpdump, which showed packets from two distinct source IPs hitting 224.1.1.1:5007 simultaneously.

YouTube Link: https://youtu.be/e0UsRd9VKa0

Individual Contributions:

- Seth Lunas: Led the implementation of Task 1 (Anycast), set up the Docker Compose configuration for both tasks, and handled tcpdump capture and logging.
- Han Htoo Zin: Led the implementation of Task 2 (Multicast), wrote the sender and receiver scripts, and managed container testing and debugging.
- Christopher Dominguez: Wrote the project report, recorded the demo video, and assisted with testing and verification across both tasks.